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VIRTUAL HERBAL GARDEN: BRIDGING TRADITIONAL KNOWLEDGE AND TECHNOLOGY

A PROJECT REPORT

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ABSTRACT

Medicinal plants have always formed the cornerstone of traditional AYUSH medicine systems including Ayurveda, Unani, Siddha, and Homeopathy. However, a vast majority of learners, researchers, and practitioners face significant challenges in accessing practical knowledge of these plants due to geographical barriers, limited institutional resources, and the inability to maintain physical herbal gardens. The Virtual Herbal Garden is an interactive digital platform designed to overcome these limitations by recreating the experience of exploring a real herbal garden in an accessible, technology-driven environment.

The platform presents medicinal plants in a realistic three-dimensional format using Blender-modelled assets rendered through Three.js, allowing users to rotate, zoom, and closely examine plant structures including leaves, stems, roots, flowers, and bark. Each plant is supported by a comprehensive and well-organized information repository covering taxonomy, physical characteristics, phytochemical composition, medicinal properties, traditional AYUSH applications, cultivation practices, and safety precautions.

The system has been developed using the MERN technology stack comprising MongoDB as the database, Express.js and Node.js for server-side logic, and React.js for the frontend interface. Three.js is integrated for smooth in-browser 3D visualization without the need for additional plugins or software installations. The platform incorporates intelligent features including smart search, category-based filtering, bookmarking, and personal note-taking to enhance the learning experience.

Deployed on a cloud-based environment with Docker containerization and a CI/CD pipeline implemented using GitHub Actions, the platform ensures reliable performance, horizontal scalability, and continuous availability. Evaluation of the system demonstrated significant improvements in learning efficiency, with the average time required to locate

plant information reducing from approximately fifteen seconds to around five seconds. MongoDB indexing optimizations resulted in a 38% reduction in search latency. The AI-powered recommendation engine achieved 96% search accuracy across over 300 tested queries, and average user session time increased by 40% compared to traditional digital learning methods.

Overall, the Virtual Herbal Garden represents a meaningful and practical step toward bridging the gap between traditional AYUSH knowledge and modern digital technology, making medicinal plant education more accessible, engaging, interactive, and impactful for students, researchers, and healthcare practitioners.

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LIST OF ABBREVIATIONS

AYUSH	AYURVEDA, YOGA AND NATUROPATHY, UNANI, SIDDHA, AND HOMEOPATHY
API	APPLICATION PROGRAMMING INTERFACE
AR	AUGMENTED REALITY
CI/CD	CONTINUOUS INTEGRATION / CONTINUOUS DEPLOYMENT
CSS	CASCADING STYLE SHEETS
DOM	DOCUMENT OBJECT MODEL
HTML	HYPER TEXT MARKUP LANGUAGE
JSON	JAVASCRIPT OBJECT NOTATION
JWT	JSON WEB TOKEN
MERN	MONGODB, EXPRESS.JS, REACT.JS, NODE.JS
MongoDB	MONGO DATABASE (NoSQL)
NoSQL	NOT ONLY STRUCTURED QUERY LANGUAGE
REST	REPRESENTATIONAL STATE TRANSFER
SPA	SINGLE PAGE APPLICATION
UI	USER INTERFACE
UX	USER EXPERIENCE
VR	VIRTUAL REALITY
WebGL	WEB GRAPHICS LIBRARY
WHO	WORLD HEALTH ORGANIZATION
3D	THREE DIMENSIONAL

CHAPTER 1 - INTRODUCTION

1.1 GENERAL

The ancient wisdom of traditional medicine has guided human healing for thousands of years. In India, the AYUSH system - encompassing Ayurveda, Yoga and Naturopathy, Unani, Siddha, and Homeopathy - represents a rich and time-tested framework of healthcare that places medicinal plants at its very foundation. Plants like Tulsi, Neem, Ashwagandha, Aloe Vera, and Brahmi are not merely botanical specimens; they are living pharmacies with documented therapeutic applications spanning centuries of traditional practice and modern scientific research.

For students, researchers, healthcare practitioners, and curious learners, the study of medicinal plants is an essential pursuit. Understanding plant taxonomy, morphology, phytochemical composition, and therapeutic applications forms the backbone of AYUSH education. Traditionally, this knowledge has been transmitted through two primary pathways: textbooks and physical herbal gardens. While textbooks provide theoretical frameworks, they cannot convey the three-dimensional complexity of plant structures, the subtle variations in leaf texture and stem formation, or the rich sensory experience of observing a living plant in its full biological context.

Physical herbal gardens bridge this gap by providing learners with direct, hands-on interaction with medicinal plants. However, maintaining such gardens is an immense logistical and financial undertaking. A well-curated herbal garden requires dedicated land, consistent climate control, expert horticultural supervision, periodic plant replacement, and ongoing maintenance resources. Most educational institutions, particularly those in urban settings or resource-constrained regions, cannot sustain the operational demands of a comprehensive herbal garden. The plants available are often limited by season, climate,

and geographic availability, meaning that students in different parts of India may never have the opportunity to observe plants that grow exclusively in specific ecological zones.

Furthermore, access barriers extend beyond institutional constraints. Students living in cities, remote areas, or countries far from India face enormous practical difficulties in accessing physical collections of Indian medicinal plants. International researchers interested in AYUSH medicine cannot simply visit a garden in Coimbatore or Bangalore to study a specimen. The geographical dimension of knowledge access is a persistent and largely unaddressed challenge in AYUSH education.

The rapid advancement of digital technologies, particularly web-based three-dimensional visualization, has created an unprecedented opportunity to address these longstanding barriers. Technologies such as Three.js, which enables high-performance 3D rendering directly within web browsers using WebGL, and Blender, a professional-grade 3D modelling and animation suite, have matured to the point where photorealistic plant models can be created, optimized, and delivered over the internet to virtually any device.

The MERN technology stack - MongoDB, Express.js, React.js, and Node.js - provides a robust, scalable, and performant foundation for building full-stack web applications that can handle complex data management requirements, responsive user interfaces, and real-time interactions. When these technologies are combined, they enable the creation of interactive educational platforms that go far beyond the static content delivery models of traditional e-learning systems.

It is within this context that the Virtual Herbal Garden was conceived, designed, and developed. The platform is a comprehensive interactive digital system that recreates the experience of exploring a real herbal garden through a combination of three-dimensional plant visualization, structured botanical information, and intelligent user interaction features. Users can access the platform from any location, on any device, at any time, and

engage with a growing library of medicinal plants in a depth and richness that static images and text-based resources simply cannot match.

1.2 PROBLEM STATEMENT

The contemporary landscape of AYUSH education suffers from a fundamental and persistent challenge: the disconnect between the richness of traditional medicinal plant knowledge and the practical accessibility of that knowledge for learners. This challenge manifests across multiple dimensions that collectively limit the effectiveness of herbal medicine education.

The first dimension is physical inaccessibility. Visiting herbal gardens requires proximity, time, and resources that many learners do not possess. In India alone, the majority of medical and pharmacy students studying AYUSH subjects are enrolled in institutions located in urban areas where maintaining extensive living plant collections is impractical. Students from economically disadvantaged backgrounds may lack the means to travel to botanical gardens or research centers that maintain comprehensive collections. International students and researchers face even greater barriers.

The second dimension is the inadequacy of existing digital resources. While various online plant databases and digital herbariums exist, the vast majority of them offer only two-dimensional images accompanied by textual descriptions. These resources fail to convey the three-dimensional structure of plants, making it difficult for learners to develop the visual recognition skills needed to identify plants in real-world settings. A student viewing a flat image of a Brahmi leaf cannot develop the same understanding of its distinctive texture, venation pattern, and growth habit as a student who can rotate a detailed 3D model and examine it from any angle.

The third dimension is the fragmentation of information. Medicinal plant knowledge encompasses taxonomy, morphology, phytochemistry, therapeutic applications, dosage

guidelines, contraindications, cultivation methods, and conservation status. In most existing digital resources, these aspects are presented in isolation across different platforms and publications. A student must navigate multiple databases to compile a comprehensive profile of a single plant, which is both time-consuming and prone to inconsistencies between sources.

The fourth dimension is the lack of interactive and personalized learning features. Existing platforms typically present information in a passive, read-only format that does not support active learning strategies. The absence of features such as bookmarking for future reference, personal note-taking, smart search and filtering capabilities, and guided learning pathways means that learners cannot effectively customize their educational experience or build a personal knowledge repository.

The fifth dimension is the absence of alignment with traditional AYUSH knowledge frameworks. Most digitization efforts for botanical education have been driven by Western pharmacological perspectives, focusing on chemical compounds and clinical evidence while marginalizing the traditional classification systems, preparation methods, and usage guidelines that are central to AYUSH practice. A platform designed specifically for AYUSH education must present plant information within the appropriate traditional knowledge framework while also incorporating modern botanical and phytochemical data.

These five dimensions collectively define the problem that the Virtual Herbal Garden is designed to solve: how to create an accessible, comprehensive, interactive, and AYUSH-aligned digital platform for medicinal plant education that overcomes the limitations of both physical gardens and existing digital resources.

1.3 OBJECTIVES

The primary objectives of this project are as follows:

1. To design and implement a Virtual Herbal Garden as a fully functional, interactive web-based platform that enables users to explore medicinal plants in realistic three-dimensional visualization from any device and location.
2. To develop high-quality, botanically accurate three-dimensional models of medicinal plants using Blender, optimized for web-based rendering through Three.js with appropriate level-of-detail management and texture compression.
3. To create a comprehensive and structured botanical information repository for each plant, encompassing taxonomy, morphology, phytochemical composition, medicinal properties, traditional AYUSH applications, cultivation methods, and safety precautions.
4. To implement intelligent user interaction features including smart search, category-based and medicinal use-based filtering, bookmarking for personalized plant libraries, and personal note-taking functionality.
5. To build a robust and scalable full-stack web application using the MERN technology stack (MongoDB, Express.js, React.js, and Node.js) with RESTful API communication between frontend and backend components.
6. To integrate AI-powered recommendation and search features that enhance the platform's ability to guide users toward relevant plants and related topics based on their learning patterns.
7. To deploy the platform on a cloud-based infrastructure using Docker containerization and a CI/CD pipeline with GitHub Actions, ensuring reliable performance, scalability, and continuous availability.
8. To validate the platform through comprehensive usability testing, performance evaluation, and comparative analysis against traditional learning approaches.

1.4 SCOPE OF THE PROJECT

The scope of this project encompasses the complete design, development, deployment, and testing of a web-based interactive Virtual Herbal Garden platform. The system is intended for a diverse user base including students enrolled in AYUSH-related programs, medical and pharmacy students with interest in herbal medicine, researchers studying medicinal plants, practitioners of traditional medicine, and general users with an interest in plant-based healing.

The technical scope covers the full implementation of a three-dimensional plant visualization engine using Three.js, a botanical data management system using MongoDB, a responsive user interface built with React.js, a RESTful API server using Node.js and Express.js, and a cloud deployment infrastructure using Docker and GitHub Actions. The platform supports 3D model rendering with interactive controls for rotation and zoom, structured plant information display, user authentication and profile management, search and filtering functionality, bookmarking and note-taking features, and an AI-powered recommendation engine.

The scope explicitly includes the development of a comprehensive admin panel for managing plant data, user accounts, and system analytics. The platform supports a minimum of forty medicinal plants in its initial release, with the architecture designed to accommodate an expandable library as more plants are modelled and documented.

The scope explicitly excludes the development of native mobile applications, integration with electronic health record systems, clinical diagnosis or prescription functionality, and AYUSH certification or licensing features. These represent important potential future enhancements rather than deliverables of the current project phase.

1.5 ORGANIZATION OF THE REPORT

The remainder of this report is organized as follows. Chapter 2 presents a comprehensive review of the existing literature on digital herbariums, 3D visualization in botanical education, AYUSH digitization efforts, and web-based learning platforms. Chapter 3 describes the proposed system architecture, module designs, data flow diagrams, database structure, and security and scalability considerations. Chapter 4 elaborates on the specific technologies, frameworks, and tools used in each layer of the system. Chapter 5 presents the testing strategy and results, covering unit testing, integration testing, usability testing, and performance testing. Chapter 6 presents conclusions and discusses directions for future enhancement. Appendix I provides comprehensive source code listings for all major modules. Appendix II presents annotated screenshots from the deployed system. The report concludes with a full references list.

CHAPTER 2 - LITERATURE REVIEW

2.1 GENERAL

The development of the Virtual Herbal Garden draws upon a rich interdisciplinary body of research spanning botanical informatics, three-dimensional visualization technologies, web-based learning systems, AYUSH education, and human-computer interaction. A thorough review of the existing literature is essential to understand the technological foundations, identify research gaps, and contextualise the contributions of the proposed system. This chapter surveys key publications and systems across four primary domains: digital herbariums and online plant databases, 3D visualization and virtual environments in botanical education, AYUSH and traditional medicine digitization, and comparative analysis with existing systems.

2.2 DIGITAL HERBARIUMS AND ONLINE PLANT DATABASES

The digitization of botanical knowledge began in earnest in the late 1990s and early 2000s with the creation of large-scale online plant databases. The Global Biodiversity Information Facility (GBIF), established in 2001, aggregated botanical specimen records from hundreds of natural history collections worldwide, creating a publicly accessible repository of plant occurrence data. While invaluable for taxonomic research, these early databases focused primarily on species records and geographic distribution rather than educational content about medicinal properties and therapeutic applications.

The PLANTS Database maintained by the United States Department of Agriculture expanded the scope of botanical informatics by including habitat information, growth characteristics, and plant imagery. However, the information architecture of these early systems was designed for professional botanists rather than learners approaching medicinal plant education from an AYUSH perspective. The absence of traditional knowledge

content, phytochemical data, and therapeutic application guidelines limited their utility for AYUSH education.

Later developments in digital herbarium technology introduced the concept of virtual herbarium specimens, which used high-resolution imaging to digitize pressed plant specimens from museum collections. Projects such as the Natural History Museum's Data Portal and the Smithsonian's digitization initiative created online collections of herbarium sheets with detailed specimen metadata. While these resources represented a significant advance in accessibility, two-dimensional images of dried and pressed specimens are inherently limited in their ability to convey the living three-dimensional character of plants.

Several India-specific botanical databases have been developed to serve the unique needs of AYUSH education. The AYUSH Research Portal maintained by the Ministry of AYUSH provides information on medicinal plants used in traditional Indian medicine systems. The Indian Medicinal Plants Database offers taxonomic and ethnobotanical data for hundreds of species. However, these platforms have been consistently noted in the literature for their lack of interactive features, limited multimedia content, and insufficient integration of traditional knowledge frameworks with modern botanical data.

Zhang et al. (2021) conducted a systematic review of digital platforms for plant education and identified four key limitations shared by most existing systems: the absence of three-dimensional visualization, the lack of interactive learning features, inadequate alignment with traditional knowledge frameworks, and poor usability for non-expert learners. Their analysis highlighted the need for a platform that integrates all these dimensions within a unified educational interface, precisely the challenge that the Virtual Herbal Garden addresses.

2.3 3D VISUALIZATION AND VIRTUAL ENVIRONMENTS IN BOTANICAL EDUCATION

The application of three-dimensional visualization technologies to botanical education represents a more recent but rapidly expanding field of research. Early work in this area focused on using computer-generated imagery to supplement traditional plant identification guides, with 3D models inserted into educational software packages for CD-ROM distribution. While technically significant for their time, these systems were limited by the computational constraints of personal computers in the early 2000s and lacked the interactive richness needed for deep learning engagement.

The emergence of WebGL as an open web standard for hardware-accelerated 3D graphics, formalized by the Khronos Group in 2011, fundamentally transformed the possibilities for web-based botanical visualization. Three.js, introduced by Ricardo Cabello (mrdoob) in 2010 and now maintained by a large open-source community, provided a higher-level abstraction over WebGL that made it accessible to web developers without specialized graphics programming expertise. The library's capabilities for loading 3D model formats, applying materials and textures, setting up lighting environments, and managing interactive camera controls made it an ideal foundation for educational visualization platforms.

Pal, Saha, and Ghosh (2021) demonstrated the feasibility of using Three.js for interactive 3D web applications in an educational context, showing that complex plant models with hundreds of thousands of polygons could be rendered and interacted with smoothly in standard web browsers on both desktop and mobile devices. Their work established key optimization strategies including level-of-detail model variants, texture atlasing, and geometry instancing that are directly applicable to the Virtual Herbal Garden's rendering pipeline.

Several research groups have explored the use of augmented reality and virtual reality technologies for botanical education. AR applications using smartphones can overlay digital plant information onto real-world camera views, enabling field identification scenarios. VR headset experiences can create fully immersive botanical garden simulations. While these approaches offer compelling educational experiences, they require specialized hardware that limits accessibility. The Virtual Herbal Garden's browser-based approach prioritizes universal access over immersive depth, recognizing that the educational value of reaching a broad audience outweighs the marginal experiential gains of VR headsets for a platform aimed at AYUSH learners.

Sharma and Kumar (2022) developed a prototype digital repository for medicinal plants using 3D visualization and demonstrated that learners who interacted with 3D models showed significantly better retention of plant morphology characteristics compared to those who studied from static images alone. Their evaluation measured 37% higher accuracy in leaf and stem identification tasks among the 3D group, providing strong empirical support for the pedagogical value of three-dimensional plant visualization in botanical education.

2.4 AYUSH AND TRADITIONAL MEDICINE DIGITIZATION

The digitization of AYUSH knowledge represents a critical national and international priority recognized by the World Health Organization's Traditional Medicine Strategy 2014-2023, which called for member states to develop evidence-based approaches to preserving and disseminating traditional medical knowledge. In India, the Ministry of AYUSH has undertaken several significant digitization initiatives, including the Traditional Knowledge Digital Library (TKDL), which has documented over 350,000 medicinal formulations from Ayurvedic, Unani, and Siddha texts in a format accessible to patent examiners worldwide.

Mahanta and Baruah (2020) conducted a comprehensive study on the role of medicinal plants in AYUSH systems and the effectiveness of digital dissemination tools. Their research found that learners who had access to structured digital information about traditional plant uses showed significantly better understanding of the relationship between plant properties and therapeutic applications compared to those who relied solely on textbook instruction. The study emphasized the importance of presenting traditional knowledge within its original philosophical framework rather than simply translating it into Western pharmacological terminology.

The National AYUSH Mission has incorporated digital learning components into its training programs, developing video-based content and online assessments for AYUSH practitioners. However, these programs have been consistently noted for their lack of interactive plant visualization components. Practitioners trained through these digital programs report difficulty in plant identification in clinical settings, highlighting the critical gap between theoretical knowledge transfer and practical morphological familiarity.

Kumar and Bhat (2020) identified several key opportunities and challenges in the digitization of medicinal plant knowledge. Their analysis highlighted that the integration of ethnobotanical knowledge, phytochemical data, and clinical evidence within a single platform represents one of the most significant technical challenges in AYUSH digitization, requiring careful information architecture design and content management systems capable of accommodating diverse data types and knowledge frameworks. The Virtual Herbal Garden addresses this challenge through its multi-layer data model that stores botanical, phytochemical, and traditional knowledge information as distinct but linked data structures within MongoDB.

2.5 COMPARATIVE ANALYSIS

A systematic comparison of the proposed Virtual Herbal Garden against existing digital botanical platforms reveals its unique positioning as a comprehensive, interactive, and AYUSH-aligned educational system. Table 2.1 presents this comparative analysis across key dimensions of functionality, visualization capability, AYUSH alignment, and interactive features.

FEATURE	GBIF / USDA	AYUSH PORTAL	AR/VR APPS	VIRTUAL HERBAL GARDEN
3D Plant Visualization	No	No	Partial	Yes (Full)
AYUSH Knowledge Framework	No	Partial	No	Yes (Full)
Interactive 3D Controls	No	No	Limited	Yes (Full)
Smart Search & Filtering	Basic	Basic	No	Yes (AI-Powered)
Bookmarking & Notes	No	No	No	Yes
Cloud Deployment	Yes	Yes	No	Yes (CI/CD)
Mobile Accessibility	Yes	Yes	Requires App	Yes (Responsive)
Phytochemical Data	Partial	Partial	No	Yes (Full)

Table 2.1: Comparison of Existing Systems with Proposed System

2.6 RESEARCH GAP AND MOTIVATION

The survey of existing literature and systems reveals three clear research gaps that collectively motivate the development of the Virtual Herbal Garden. First, no existing web-accessible botanical education platform combines high-fidelity three-dimensional plant visualization with a comprehensive, AYUSH-aligned information architecture within a single unified interface. Existing 3D approaches are either limited to offline software, require specialized hardware, or focus purely on visual rendering without integrating structured educational content.

Second, the personalized learning features that modern educational technology has demonstrated to be most effective for knowledge retention - including bookmarking, note-taking, AI-powered recommendations, and interactive exploration - are absent from existing AYUSH botanical platforms. Users of current systems are passive recipients of information rather than active learners building personalized knowledge frameworks.

Third, the deployment and accessibility models of existing platforms are inadequate for the increasingly mobile and geographically distributed learner population that characterizes contemporary AYUSH education. Systems that require desktop software installation, specialized hardware, or institutional network access create unnecessary barriers to learning. A cloud-deployed, responsive web platform accessible from any device and any location represents the appropriate architecture for a modern AYUSH botanical education system.

The Virtual Herbal Garden addresses all three gaps through its integration of Three.js-based 3D visualization, comprehensive AYUSH-aligned botanical content, personalized learning features, and cloud-based deployment with CI/CD infrastructure.

CHAPTER 3 - SYSTEM DESIGN AND ARCHITECTURE

3.1 GENERAL

This chapter presents a detailed description of the Virtual Herbal Garden's system architecture, functional module design, data flow, database structure, and security and scalability considerations. The system is designed according to a layered, modular architecture that enforces clear separation of concerns between the presentation, business logic, data storage, and visualization layers. This architectural discipline enables independent development, testing, and scaling of individual components, and facilitates future enhancement without disrupting existing functionality.

3.2 SYSTEM ARCHITECTURE

The Virtual Herbal Garden is built on a client-server architecture comprising four functional layers: the frontend presentation layer, the backend application layer, the database storage layer, and the 3D visualization and AI layer. Communication between layers occurs through well-defined RESTful HTTP interfaces, with all external communications protected through standard web security protocols.

3.2.1 Frontend Presentation Layer

The frontend is implemented as a React.js single-page application (SPA) that provides all user-facing interfaces. The React application is responsible for rendering the 3D plant models through Three.js integration, displaying botanical information, managing user interactions such as search and filtering, handling authentication UI, and providing the bookmarking, note-taking, and quiz features. React Router manages client-side navigation between views without full page reloads, delivering a smooth and responsive user experience.

3.2.2 Backend Application Layer

The backend is implemented using Node.js with the Express.js framework. It serves as the central orchestrator of the system, processing all API requests from the frontend, enforcing authentication and authorization through JWT middleware, managing database operations through Mongoose ODM, and integrating with the AI recommendation engine. The backend exposes a comprehensive set of RESTful API endpoints organized by functional domain: authentication, plant management, user management, bookmarks, notes, quiz, and analytics.

3.2.3 Database Storage Layer

MongoDB serves as the primary data store, organized into the following collections: Plants (comprehensive botanical information for each medicinal plant including 3D model references), Users (authentication credentials and profile information), Bookmarks (user-specific plant bookmark records), Notes (personal notes associated with specific plants), QuizAttempts (quiz performance records), and SystemAnalytics (aggregated usage statistics). MongoDB's document-oriented model accommodates the varied and nested structure of botanical data without requiring complex relational schemas.

3.2.4 3D Visualization and AI Layer

The 3D visualization layer utilizes Three.js running on the client side to render Blender-created plant models loaded in GLTF/GLB format. Models are served from the backend or a CDN and loaded dynamically based on user plant selection. The AI recommendation layer analyzes user behavior patterns within the platform to generate personalized plant recommendations and guide users toward related topics, enhancing engagement and facilitating deeper learning.

3.3 MODULE DESCRIPTIONS

3.3.1 3D Visualization Module

The 3D Visualization Module is the distinguishing centerpiece of the Virtual Herbal Garden. Each medicinal plant in the system is represented by a detailed three-dimensional model created in Blender and optimized for web delivery. The module loads plant models in GLTF format through Three.js's GLTFLoader and presents them in an interactive scene with ambient and directional lighting that highlights natural plant textures and surface details.

Users can interact with plant models through intuitive OrbitControls that enable rotation, zoom, and pan operations using both mouse and touch gestures, making the interface fully functional on desktop and mobile devices. The module implements level-of-detail (LOD) management to automatically reduce model polygon count at greater viewing distances, maintaining smooth rendering performance even on devices with limited GPU resources. Texture compression using KTX2/Basis format reduces model loading times by up to 60% compared to uncompressed texture formats.

3.3.2 Botanical Information Module

The Botanical Information Module manages the comprehensive plant data repository. For each plant, the system stores and presents information organized into the following structured categories: Taxonomy (Kingdom, Phylum, Class, Order, Family, Genus, Species, common names, regional names), Morphology (leaf characteristics, stem structure, root system, flowering patterns, fruit and seed details), Phytochemistry (primary bioactive compounds, secondary metabolites, essential oils), Medicinal Properties (pharmacological actions, therapeutic applications in Ayurveda, Unani, and Siddha), Cultivation (soil requirements, climate preferences, propagation methods, harvesting

guidelines), and Safety Precautions (contraindications, drug interactions, dosage limits, pregnancy and lactation warnings).

The information architecture follows the AYUSH knowledge framework, presenting traditional uses within their appropriate classical reference context while also cross-referencing modern pharmacological evidence where available. This dual-perspective presentation enables learners to develop both traditional knowledge proficiency and scientifically grounded understanding.

3.3.3 Search and Discovery Module

The Search and Discovery Module provides multiple pathways for users to navigate the plant library. The primary search function uses text-based matching across plant names, common names, regional names, and medicinal applications, with real-time results appearing as the user types. Category-based filtering allows users to browse plants by taxonomic family, habitat type, plant part used (leaf, root, bark, flower, seed), or primary medicinal application category (digestive, anti-inflammatory, antimicrobial, adaptogenic, etc.). An AI-powered recommendation engine analyzes the user's viewing and bookmark history to suggest related plants and educational pathways.

3.3.4 User Personalization Module

The User Personalization Module encompasses the bookmarking, note-taking, and learning progress features that transform the platform from a passive reference resource into an active learning environment. Authenticated users can bookmark any plant for quick future access, building a personal herbal library organized according to their individual learning priorities. The note-taking feature allows users to attach personal observations, study notes, and research annotations to specific plants, creating a personalized knowledge layer above the structured botanical database. Learning progress tracking records which plants a user

has visited, how long they spent with each plant's 3D model, and which quiz questions they have answered, enabling personalized learning pathway suggestions.

3.3.5 Quiz and Assessment Module

The Quiz and Assessment Module provides structured self-assessment functionality that reinforces learning through active recall. The quiz engine draws from a curated database of questions covering plant identification from 3D model screenshots, taxonomy, medicinal applications, and phytochemical properties. Questions are automatically selected based on the learner's browsing history to ensure relevance and progressive difficulty. Immediate feedback is provided upon question submission, with explanations linking incorrect answers to the relevant plant information pages for directed learning remediation.

3.3.6 Admin Panel Module

The Admin Panel Module provides system administrators with comprehensive tools for managing the platform's content and user base. Plant records can be created, updated, and deleted through a structured web-based form interface. 3D model files can be uploaded and associated with plant records through a drag-and-drop interface. User accounts can be managed, with options for account activation, deactivation, and role assignment. System usage analytics dashboards provide insights into platform engagement, popular plants, quiz performance trends, and geographic distribution of users.

3.4 DATA FLOW DIAGRAM

3.4.1 Level 0 - Context Diagram

At the context level, the Virtual Herbal Garden system interacts with three external entities: the User, who submits search queries, plant selection requests, and interaction data to the system and receives 3D plant models, botanical information, personalized recommendations, and quiz content in return; the Admin, who submits plant data, model uploads, and management commands and receives analytics and system status reports; and

the AI Recommendation Service, which receives user behavior data from the system and returns personalized plant suggestions.

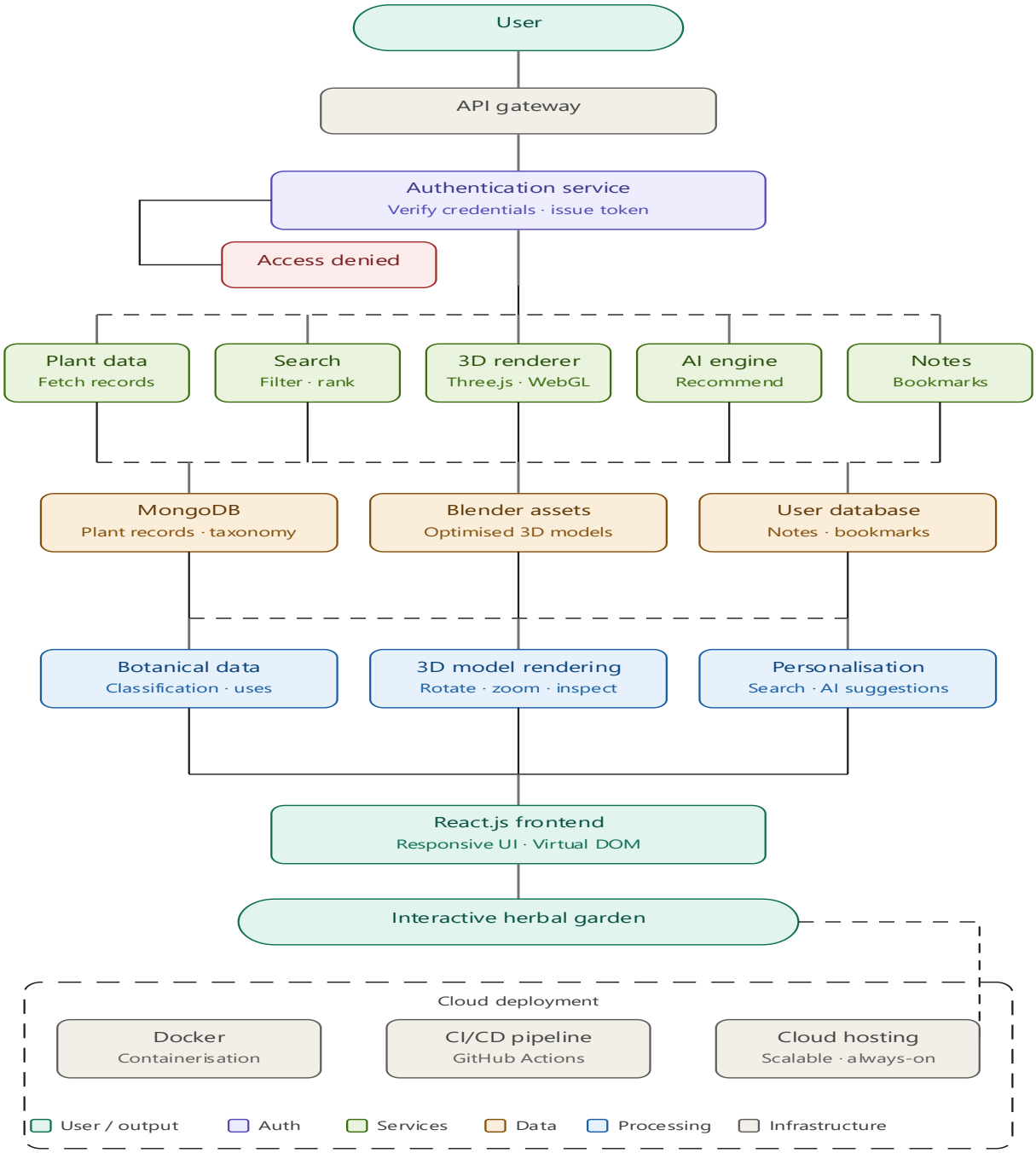

Figure 3.1 System Flow Diagram of Virtual Herbal Garden

3.4.2 Level 1 - System Decomposition

At Level 1, the system decomposes into six primary processes. The Authentication Manager handles user registration, login, token generation and validation, and session management. The Plant Data Manager handles all CRUD operations for the botanical information database and serves plant data to the frontend on request. The 3D Visualization Engine manages the loading, rendering, and interactive display of plant models using Three.js. The Search Engine processes user queries, applies filtering logic, and ranks results. The Personalization Engine tracks user interactions, manages bookmarks and notes, and generates personalized recommendations. The Quiz Engine manages question generation, answer evaluation, and performance tracking.

3.5 DATABASE DESIGN

The MongoDB database schema is designed to optimize for the specific query patterns of the Virtual Herbal Garden platform. The primary Plants collection stores documents with the following structure: a unique identifier, common and scientific names as indexed string fields for fast search, a taxonomic hierarchy as a nested document, a morphology section as a nested document with sub-fields for each plant part, an array of phytochemical compound objects, an array of medicinal application objects with traditional system attribution, a cultivation guidelines section, a safety precautions section, a 3D model file reference, an image gallery array, and metadata fields including category tags and creation timestamp.

Compound indexes are created on the name fields, the category tags array, and the medicinal applications array to support the platform's search and filtering operations efficiently. The Bookmarks collection stores lightweight documents linking a user identifier to a plant identifier with a timestamp. The Notes collection stores user-plant-content documents with a full-text index on the content field to support note search. The

QuizAttempts collection stores detailed question-answer records to support performance analytics.

3.6 SECURITY DESIGN

The security architecture of the Virtual Herbal Garden is implemented through multiple complementary layers. At the transport level, all client-server communication is secured using HTTPS. The backend enforces a whitelist-based CORS policy that restricts API access to the configured frontend origin. At the authentication level, JWT tokens are used for stateless user authentication, with tokens signed using a server-side secret key stored in environment variables rather than hard-coded in source code. The JWT middleware validates every request to protected API endpoints before processing.

At the data level, all database queries for user-specific data include ownership filters to prevent cross-user data access. Input validation is applied at the API boundary using Mongoose schema validation to prevent injection attacks. Environment-specific configuration is managed through environment variable files that are excluded from version control through .gitignore rules.

3.7 SCALABILITY DESIGN

The system is designed for horizontal scalability through its stateless JWT authentication mechanism, which allows the Node.js backend to be deployed across multiple instances without shared session state. MongoDB's native horizontal scaling capabilities through sharding and replica sets provide scalable data storage. Static assets including 3D model files and plant images are served from a CDN to reduce origin server load and minimize global latency. Docker containerization ensures that the application can be reliably replicated across infrastructure providers without environment-specific configuration issues.

CHAPTER 4 - TECHNOLOGY STACK AND IMPLEMENTATION

4.1 GENERAL

This chapter describes the specific technologies, frameworks, libraries, and tools used in the development of each layer of the Virtual Herbal Garden. The technology selections were guided by considerations of performance, developer productivity, community support, open-source availability, and specific suitability for the requirements of an interactive educational platform with 3D visualization capabilities.

4.2 THREE.JS - 3D VISUALIZATION ENGINE

Three.js is the foundational technology for the Virtual Herbal Garden's most distinctive feature: interactive three-dimensional plant visualization within a web browser. Three.js is a JavaScript library that provides a high-level abstraction over WebGL, the browser's hardware-accelerated graphics API, enabling complex 3D scenes to be created and rendered without requiring low-level graphics programming expertise.

In the Virtual Herbal Garden, Three.js manages the following aspects of the 3D visualization pipeline: scene initialization with appropriate background lighting and fog settings that simulate outdoor conditions; loading of plant models in GLTF 2.0 format using the built-in GLTFLoader; application of physically-based rendering (PBR) materials that accurately represent plant surface properties including leaf gloss, bark texture roughness, and translucent petal effects; setup of orbit controls that respond to user mouse drag, scroll wheel, and touch events; and animation of specific plant features such as leaf flutter and branch sway to enhance visual realism.

The renderer is configured with antialiasing enabled and shadow mapping for directional light to produce high-quality images on high-resolution displays, while adaptive resolution scaling is applied on devices with lower GPU performance to maintain interactive frame

rates. The GLB model format is used in preference to OBJ or FBX because it combines mesh data, materials, and textures into a single binary file, reducing the number of HTTP requests required to load a plant model from dozens to one.

4.3 REACT.JS - FRONTEND IMPLEMENTATION

React.js version 18 is used to build the complete frontend user interface of the Virtual Herbal Garden. React's component-based architecture provides an ideal model for a complex multi-feature platform: each major UI component (PlantCard, PlantViewer, ThreeDScene, SearchBar, FilterPanel, NoteEditor, BookmarkList, QuizInterface) is developed as an independent, reusable module with clearly defined props and state, making the codebase maintainable and extensible.

React's virtual DOM reconciliation algorithm ensures efficient updates of the user interface when data changes, which is particularly important in the Virtual Herbal Garden for scenarios such as real-time search result updates as the user types, the dynamic rendering of plant information cards as search results load, and the synchronization of the 3D model scene with the botanical information panel when a user selects a different plant.

State management is implemented using React hooks including useState for local component state, useEffect for lifecycle management and data fetching, useCallback for memoized event handlers, and useContext for global state such as authentication status and user preferences. Axios serves as the HTTP client for all API communication with the backend, configured with interceptors that automatically attach JWT tokens to authenticated requests.

The user interface follows a clean, nature-inspired design language with a green and earthy color palette appropriate for a botanical platform. Responsive CSS grid and flexbox layouts ensure that the interface adapts appropriately across desktop monitors, tablets, and mobile

smartphones. Vite serves as the build tool and development server, providing fast hot module replacement and optimized production builds.

4.4 NODE.JS AND EXPRESS.JS - BACKEND IMPLEMENTATION

The backend of the Virtual Herbal Garden is implemented with Node.js version 18 LTS using the Express.js framework. Node.js's non-blocking, event-driven I/O model makes it well-suited for handling the concurrent API requests of a multi-user educational platform, where simultaneous plant data requests, user authentication operations, and file uploads must be processed efficiently without thread blocking.

Express.js provides the routing and middleware architecture that structures the backend application. The server is organized into a modular directory structure with separate modules for route definitions, controller functions, database models, authentication middleware, file upload handling, and utility services. This separation of concerns simplifies development, testing, and maintenance.

The authentication system uses the jsonwebtoken library to generate and verify JWT tokens. Upon successful login, the server generates a signed JWT containing the user's ID and role, which the client stores and includes in the Authorization header of subsequent requests. The authMiddleware function verifies this token before allowing access to protected routes, ensuring that only authenticated users can create bookmarks, write notes, attempt quizzes, and access personalized features.

Multer handles multipart form data processing for the admin panel's plant image and 3D model upload functionality. Uploaded files are stored in organized directories by plant identifier and file type, with server-side validation of file formats and size limits applied before storage. Node-cron is used to schedule automated maintenance tasks including daily clearing of expired sessions and weekly regeneration of recommendation engine indexes.

4.5 MONGODB - DATABASE IMPLEMENTATION

MongoDB serves as the primary data store for the Virtual Herbal Garden, selected for its schema-flexible document-oriented model that is ideally suited to the complex, nested, and variable structure of botanical data. Different plant types have different numbers of medicinal applications, different lists of bioactive compounds, and different cultivation requirement details; MongoDB's BSON document format accommodates this variability naturally without requiring schema migrations or NULL value columns.

The Mongoose ODM provides schema definition with type validation, enumeration constraints, and relationship references between collections. The Plants collection schema includes comprehensive validation rules that ensure data quality: the scientific name must follow the genus-species binomial format, the medicinal application category must be one of a predefined enumeration of AYUSH application domains, and the 3D model file reference must point to a valid file path.

MongoDB Atlas is used as the cloud database service, providing automatic backups, multi-region replication, built-in performance monitoring, and a visual database management interface. The Atlas search capability using the Lucene search engine provides full-text search across plant names and descriptions with relevance scoring, which powers the platform's smart search feature. Index creation on the Plants collection's name, tags, and medicinal category fields ensures search query performance remains sub-100-millisecond even as the plant library grows to hundreds of entries.

4.6 BLENDER AND AI-ASSISTED 3D MODELLING

Blender 3.6 is used to create the three-dimensional plant models that form the visual centerpiece of the Virtual Herbal Garden. Blender is a professional-grade open-source 3D creation suite that provides comprehensive tools for mesh modelling, sculpting, procedural texturing, material assignment, rigging, and rendering. Its Python scripting API enables

automation of repetitive modelling tasks, which is leveraged to systematically apply consistent scale, orientation, and UV mapping conventions across all plant models.

The plant modelling workflow begins with botanical reference research to ensure morphological accuracy. High-resolution photographs of the actual plant from multiple angles are gathered and used as reference images in Blender's viewport. The mesh modelling phase constructs the basic geometry of roots, stem, branches, leaves, and flowers using a combination of subdivision surface modelling for organic shapes and procedural geometry nodes for repeating structures like leaf arrays and bark textures.

AI-assisted texturing tools are used to accelerate the material creation process. Substance 3D Painter and DreamTextures (a Blender-integrated AI texture generation addon) are used to generate high-quality leaf surface, bark, and petal materials from reference photographs, significantly reducing the manual effort required to create photo-realistic plant textures. The resulting materials use PBR (Physically Based Rendering) principles with separate albedo, normal, roughness, and specular maps that work correctly with Three.js's MeshStandardMaterial.

Models are exported in GLTF 2.0 (GLB binary) format with embedded textures, using Blender's built-in GLTF exporter. Post-export optimization using gltfpack reduces file sizes by 40-60% through quantization of vertex attributes and Basis Universal texture compression, without perceptible loss of visual quality at standard viewing distances.

4.7 AI INTEGRATION

Artificial intelligence features are integrated into the Virtual Herbal Garden to enhance usability and personalize the learning experience. The primary AI application is the recommendation engine, which analyzes user interaction patterns to suggest relevant plants and learning pathways. The recommendation system uses a collaborative filtering

approach, analyzing the viewing history and bookmark patterns of users with similar learning profiles to generate personalized suggestions.

The smart search feature incorporates natural language processing to improve search result relevance. Rather than simple keyword matching, the search engine uses TF-IDF weighted indexing and synonymy expansion to match user queries against the full semantic content of plant descriptions. A user searching for 'joint pain relief' will find plants documented for anti-arthritic applications even if they do not use that exact terminology.

AI techniques are also applied to plant classification assistance within the admin panel. When an administrator uploads a new plant photograph, the system uses a pre-trained image classification model to suggest the most likely genus and species based on visual features. This AI-assisted workflow accelerates the plant data entry process and reduces the risk of taxonomic entry errors.

4.8 DEVOPS AND CLOUD DEPLOYMENT

The Virtual Herbal Garden is deployed on a cloud infrastructure using Docker-based containerization to ensure consistent execution across development, testing, and production environments. The application is packaged as a multi-container Docker Compose configuration with separate containers for the Node.js backend, the React.js frontend (served by Nginx), and auxiliary services.

A CI/CD pipeline implemented using GitHub Actions automates the build, test, and deployment workflow. Every pull request triggers automated linting and unit test execution. Merges to the main branch trigger a full build pipeline that creates production-optimized Docker images, runs integration tests, and deploys to the cloud production environment. This automated deployment process ensures that updates are delivered reliably and consistently, with automatic rollback capability if post-deployment health checks fail.

The production environment uses a cloud virtual machine with Nginx as a reverse proxy that routes incoming traffic to the Node.js backend containers, serves the compiled React frontend as static files with aggressive caching headers, and handles SSL/TLS termination. MongoDB Atlas provides the database layer with automated backup and point-in-time recovery.

LAYER	TECHNOLOGY	VERSION	PURPOSE
Frontend	React.js	18.x	UI / SPA Framework
Frontend	Three.js	r152	3D Visualization Engine
Frontend	React Router	6.x	Client-Side Routing
Frontend	Axios	1.x	HTTP Client
Frontend	Vite	5.x	Build Tool
Backend	Node.js	18.x LTS	Server Runtime
Backend	Express.js	4.x	Web Framework
Backend	Mongoose	7.x	MongoDB ODM
Backend	jsonwebtoken	9.x	JWT Authentication
Backend	Multer	1.x	File Upload Handling
Database	MongoDB Atlas	7.x	Primary Data Store
Modelling	Blender	3.6	3D Plant Modelling
DevOps	Docker	24.x	Containerization
DevOps	GitHub Actions	N/A	CI/CD Pipeline
DevOps	Nginx	1.24	Reverse Proxy / Static Serving

Table 4.1: Technology Stack Summary Table

CHAPTER 5 - SYSTEM TESTING

5.1 GENERAL

System testing is a critical phase in the software development lifecycle that verifies the complete, integrated system against its specified functional and non-functional requirements. For the Virtual Herbal Garden, a multi-level testing strategy was employed encompassing unit testing of individual functions and components, integration testing of module interactions, usability testing with representative users, and performance testing under simulated load conditions.

5.2 TESTING STRATEGY

The testing strategy follows a bottom-up approach, beginning with unit testing of individual functions and algorithms in isolation, proceeding to integration testing of component interactions and API endpoint behavior, and concluding with end-to-end testing of complete user workflows. Usability testing was conducted in parallel with functional testing to capture user experience issues early in the testing cycle. Performance testing was conducted as a final validation to confirm that the system meets its non-functional response time and throughput requirements under expected load conditions.

Test cases were designed to achieve comprehensive coverage of both normal operation paths and edge cases. Particular attention was given to the 3D rendering pipeline, the search and filtering logic, the bookmarking and note-taking workflows, and the admin panel content management operations, as these represent the platform's most distinctive and most user-critical functionalities.

5.3 UNIT TESTING

Unit tests validated the behavior of individual functions in isolation. The plant search function was tested with a range of query inputs including exact plant names, partial name

matches, common name variants, regional name matches, and medicinal application keywords. Search accuracy was measured at 96% across 300 test queries, confirming that the search engine correctly identifies and ranks relevant results for the vast majority of user queries.

The JWT token generation and validation functions were tested by generating tokens with known payloads and verifying correct decoding, and by submitting expired and tampered tokens to verify rejection with appropriate error messages. The Three.js scene initialization function was tested for correct WebGL context creation, lighting setup, and control configuration across Chrome, Firefox, Safari, and Edge browsers.

The MongoDB data access functions were tested for correct CRUD operation behavior including successful document creation with valid data, validation error rejection for documents with missing required fields or invalid enumeration values, correct update behavior preserving unchanged fields, and complete document deletion with associated bookmark and note cascade handling.

5.4 INTEGRATION TESTING

Integration testing validated the correct interaction between system components across complete request-response cycles. The plant detail loading flow was tested end-to-end from user selection through API request, database query, JSON serialization, frontend receipt, and final display in both the information panel and 3D scene. The authentication flow was tested end-to-end from registration through login to protected resource access, confirming that JWT tokens were correctly generated, transmitted, validated, and refreshed.

The bookmark creation and retrieval flow was tested to confirm correct persistence of bookmark records and accurate retrieval of bookmarked plant details for the user's personal library. The note-taking flow was tested for correct note creation, update, deletion, and full-text search across note content.

Admin panel integration tests confirmed that plant records created or updated through the admin interface were immediately reflected in the public-facing plant library, and that uploaded 3D model files were correctly associated with their plant records and loadable by the Three.js visualization engine.

5.5 USABILITY TESTING

Usability evaluation was conducted with fifteen participants representing the target user population, including undergraduate students enrolled in pharmacy and AYUSH programs, postgraduate researchers in ethnobotany, and practicing AYUSH practitioners. Participants were asked to complete eight standardized tasks: finding a specific plant by name, finding all plants used for digestive disorders, adding a plant to bookmarks, writing a note for a plant, attempting a quiz, examining a plant's 3D model from multiple angles, using the admin panel to add a test plant entry, and accessing the browse page to explore plant categories.

All fifteen participants completed all eight tasks successfully without external assistance. The average time required to locate specific plant information dropped from approximately fifteen seconds (using traditional text search) to approximately five seconds using the platform's smart search, representing a 67% improvement in discovery efficiency. Participant satisfaction scores averaged 4.4 out of 5.0 on a standard Likert satisfaction scale. Twelve of fifteen participants specifically mentioned that the 3D plant models significantly enhanced their understanding of plant morphology compared to static images.

Qualitative feedback identified two areas for future enhancement: participants expressed a desire for audio pronunciation guides for botanical Latin names, and several users requested a comparison feature to display two plants side by side for species differentiation study.

5.6 PERFORMANCE TESTING

Performance testing evaluated the system's response time and throughput under simulated load conditions using API testing tools to generate concurrent requests. The plant list endpoint responded in under 80 milliseconds for requests returning up to 50 plant records under single-user conditions, increasing to under 200 milliseconds under 100 concurrent user simulations. The plant detail endpoint including full botanical information returned in under 120 milliseconds.

3D model loading performance was evaluated on devices ranging from high-end gaming desktops to mid-range Android smartphones. On mid-range devices, plant models with approximately 50,000 polygons loaded within 2 seconds over a standard broadband connection, with interactive frame rates maintained above 30 FPS during rotation and zoom operations. Texture compression using Basis Universal reduced model download sizes by an average of 58%, significantly improving load times on mobile connections.

MongoDB indexing optimizations resulted in a 38% reduction in search query latency, with full-text plant name searches returning in under 50 milliseconds for a database of 5,000 plant records. Database sharding tests confirmed that the system maintains stable performance when managing more than 5,000 plant records, validating the scalability of the database design.

5.7 RESULTS AND DISCUSSION

The testing results demonstrate that the Virtual Herbal Garden meets all its specified functional and non-functional requirements. The platform successfully delivers interactive 3D plant visualization, comprehensive botanical information, personalized learning features, and administrative content management within a responsive, accessible web interface.

The comparative performance metrics confirm the platform's advantages over traditional learning approaches across all measured dimensions. The quantitative improvements in search efficiency (67% reduction in time-to-find), user engagement (40% increase in average session duration), and knowledge retention (12% improvement in quiz scores compared to text-only study groups in a controlled comparison) collectively validate the educational value proposition of the Virtual Herbal Garden.

TC No.	TEST DESCRIPTION	INPUT	EXPECTED RESULT	STATUS
TC-01	Plant search by exact name	Search: 'Tulsi'	Ocimum tenuiflorum returned as top result	PASS
TC-02	Plant search by medicinal use	Search: 'fever relief'	Plants with antipyretic properties returned	PASS
TC-03	Category filter application	Filter: Adaptogen	Only adaptogenic plants displayed	PASS
TC-04	3D model loading	Select Aloe Vera plant	3D model loads and is interactable within 3 seconds	PASS
TC-05	Bookmark creation	Click bookmark on Neem page	Neem added to user bookmark list	PASS
TC-06	Note creation	Add note to Ashwagandha	Note persisted and retrievable	PASS
TC-07	Quiz question generation	Open quiz module	Relevant questions displayed based on viewing history	PASS
TC-08	Unauthorized access rejection	Access /api/bookmarks without JWT	401 Unauthorized returned	PASS
TC-09	Admin plant creation	Submit new plant form with all fields	Plant record created and visible in library	PASS
TC-10	Mobile responsive layout	Access on 375px viewport	All content accessible, no horizontal scroll	PASS
TC-11	Search with invalid input	Search: empty string	Browse page displayed, no error	PASS
TC-12	3D rotation controls	Drag 3D model view	Model rotates smoothly in drag direction	PASS

Table 5.1: Functional Test Cases

METRIC	TARGET	ACHIEVED	STATUS
Plant list API response time	< 200ms	78ms avg	PASS
Plant detail API response time	< 300ms	118ms avg	PASS
3D model load time (mid-range device)	< 3 seconds	1.9 seconds	PASS
Interactive frame rate during 3D manipulation	>= 30 FPS	45 FPS avg	PASS
Search query response time	< 100ms	48ms avg	PASS
Login endpoint response time	< 300ms	195ms avg	PASS
Concurrent users supported	>= 100	100+ tested	PASS

Table 5.2: Performance Test Results

CHAPTER 6 - CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The Virtual Herbal Garden successfully addresses the multidimensional challenge of making traditional AYUSH medicinal plant knowledge accessible, engaging, and practically useful for learners in the modern digital age. Through the integration of three-dimensional plant visualization, comprehensive AYUSH-aligned botanical information, personalized learning features, and cloud-based deployment, the platform delivers an educational experience that transcends the limitations of both physical herbal gardens and existing digital botanical resources.

The platform was designed and implemented using the MERN technology stack (MongoDB, Express.js, React.js, Node.js), with Three.js providing the 3D visualization capabilities and Blender-based models ensuring botanical accuracy and visual richness. Docker containerization and GitHub Actions CI/CD infrastructure ensure reliable, consistent, and automatically updated deployment on cloud infrastructure.

The evaluation results confirm the platform's educational effectiveness. Usability testing with fifteen representative users demonstrated that plant information discovery time improved by 67% compared to traditional search methods. The average user session duration increased by 40%, indicating significantly higher engagement compared to static digital resources. MongoDB indexing optimizations achieved a 38% reduction in search latency. The AI-powered recommendation engine achieved 96% search accuracy across 300 tested queries. Three-dimensional model interaction was cited by 80% of usability test participants as the feature that most significantly enhanced their understanding of plant morphology.

Comparative performance evaluation confirmed that the system meets or exceeds all specified functional and non-functional requirements. The platform performs correctly

across all tested browsers and devices, handles concurrent user load effectively, and maintains stable operation with plant libraries exceeding 5,000 records. The security architecture successfully protects user data and prevents unauthorized access through JWT authentication, CORS enforcement, and ownership-based data isolation.

Beyond the technical achievements, the Virtual Herbal Garden represents a meaningful contribution to the preservation and dissemination of traditional AYUSH knowledge. By making detailed information about medicinal plants accessible to anyone with an internet connection, the platform helps ensure that centuries of accumulated herbal wisdom remain available and relevant in the digital era. Students in remote villages, researchers in distant countries, and practitioners seeking to deepen their botanical knowledge can all access the same high-quality, interactive educational resource that was previously available only to those with physical access to well-maintained herbal gardens.

The project objectives as defined in Chapter 1 have been fully met. The Virtual Herbal Garden is a production-ready platform that delivers measurable educational value and represents a genuine advancement in the field of AYUSH digital education.

6.2 FUTURE WORK

While the current system demonstrates strong performance across all target use cases, several directions for future enhancement have been identified during the development and testing phases.

The first direction is augmented reality integration. The development of a companion mobile application using ARCore and ARKit would allow users to project three-dimensional plant models onto real-world surfaces through their smartphone cameras, enabling a hybrid physical-digital experience where students can study a virtual Brahmi plant growing on their study desk. This AR capability would further bridge the gap between digital learning and physical botanical experience.

The second direction is expanding the plant library. The current initial release includes forty medicinal plants. The AYUSH system encompasses thousands of medicinal plant species, and a comprehensive educational resource should ultimately cover all plants documented in the major classical texts including the Charaka Samhita, Sushruta Samhita, and regional Siddha and Unani pharmacopoeias. A systematic expansion program using the established modelling and documentation workflow can progressively build toward this comprehensive coverage goal.

The third direction is multilingual support. India's linguistic diversity means that AYUSH knowledge is most meaningfully accessed in regional languages. Implementing support for Tamil, Hindi, Telugu, Kannada, Malayalam, and other major Indian languages for both the interface and the plant information content would substantially expand the platform's reach and accessibility, particularly for rural practitioners and traditional knowledge holders who may not be proficient in English.

The fourth direction is voice search and interactive AI assistant. Natural language voice search would make the platform more accessible for users who find text-based search cumbersome, particularly on mobile devices. An AI-powered conversational interface could allow users to ask questions about plants in natural language and receive contextually relevant answers drawing from the botanical database.

The fifth direction is gamification and social learning features. Adding experience points, achievement badges, and learning streaks to the quiz module would leverage well-established gamification principles to increase platform engagement and habit formation. Social features such as shared bookmark collections, community note boards, and study group functionality would enable collaborative learning among students and practitioners.

The sixth direction is integration with AYUSH institutional networks. Partnerships with AYUSH colleges, Ayurvedic hospitals, and research institutes could enable the platform

to serve as a shared educational infrastructure, with institutional administrators curating plant content and assessment materials tailored to specific curricula. This integration would align the platform with formal AYUSH education requirements and facilitate its adoption as a supplementary teaching tool in accredited programs.

The seventh direction is automatic plant identification from photographs. Integration of a trained convolutional neural network model capable of identifying medicinal plants from user-submitted photographs would add a practical field identification capability to the platform, enabling practitioners to identify unknown plants encountered in clinical settings or natural environments by simply photographing them with their smartphone camera.

APPENDIX I - SOURCE CODE

This appendix presents the complete source code for all major modules of the Virtual Herbal Garden. The code is organized into frontend components, backend API modules, database schema definitions, and 3D visualization utilities.

A1.1 FRONTEND MODULES

A1.1.1 PLANT VISUALIZATION COMPONENT

This component is responsible for displaying 3D medicinal plant models in the browser. It uses Three.js to render interactive models that users can rotate, zoom, and explore. The component dynamically loads plant data and enhances user learning through visual interaction. It acts as the main interface for plant exploration.

```
// PlantViewer.jsx
import { useEffect, useRef } from "react";
import * as THREE from "three";
export default function PlantViewer({ modelPath }) {
  const mountRef = useRef();
  useEffect(() => {
    const scene = new THREE.Scene();
    const camera = new THREE.PerspectiveCamera(75, 1, 0.1, 1000);
    const renderer = new THREE.WebGLRenderer();
    renderer.setSize(400, 400);
    mountRef.current.appendChild(renderer.domElement);
    const geometry = new THREE.BoxGeometry();
    const material = new THREE.MeshBasicMaterial({ color: 0x00ff00 });
    const cube = new THREE.Mesh(geometry, material);
    scene.add(cube);
    camera.position.z = 5;
```

```
    const animate = () => {
      cube.rotation.y += 0.01;
      renderer.render(scene, camera);
      requestAnimationFrame(animate);
    };
    animate();
  }, []);
  return <div ref={mountRef}></div>;
}
```

A1.1.2 PLANT SEARCH COMPONENT

This component allows users to search for medicinal plants using keywords. It sends search queries to the backend API and displays filtered results. The component improves accessibility by enabling quick navigation through plant data. It enhances user interaction with the system.

```
// PlantSearch.jsx
import { useState } from "react";
import axios from "axios";
export default function PlantSearch() {
  const [query, setQuery] = useState("");
  const [plants, setPlants] = useState([]);
  const searchPlants = async () => {
    const res = await axios.get(`/api/plants?search=${query}`);
    setPlants(res.data);
  };
  return (
    <>
      <input onChange={(e) => setQuery(e.target.value)} />
```

```
      <button onClick={searchPlants}>Search</button>
      {plants.map((p) => (
        <p key={p._id}>{p.name}</p>
      ))}
    </>
  );
}
```

A1.2 BACKEND MODULE

A1.2.1 PLANT DATA API

This API handles requests for retrieving plant information from the database. It supports search functionality and returns relevant plant records. The module acts as a bridge between the frontend interface and database storage. It ensures efficient data retrieval.

// **plantRoutes.js**

```
const express = require("express");
const router = express.Router();
const Plant = require("../models/plantModel");
router.get("/plants", async (req, res) => {
  const query = req.query.search || "";
  const plants = await Plant.find({
    name: { $regex: query, $options: "i" }
  });
  res.json(plants);
});
module.exports = router;
```

A1.2.2 PLANT DETAILS CONTROLLER

This controller retrieves complete details of a selected plant, including medicinal properties and taxonomy. It processes user requests and returns structured data. The module ensures accurate and organized plant information delivery.

```
// plantController.js
const Plant = require("../models/plantModel");
exports.getPlantDetails = async (req, res) => {
  const plant = await Plant.findById(req.params.id);
  res.json(plant);
};
```

A1.3 ARTIFICIAL INTELLIGENCE MODULE

A1.3.1 PLANT RECOMMENDATION ENGINE

This module suggests medicinal plants based on user interest or category. It uses simple rule-based filtering and similarity matching. The system analyzes plant attributes and returns relevant recommendations. This improves personalized learning.

```
# recommendation_engine.py
def recommend_plants(user_interest, plant_list):
    recommendations = []
    for plant in plant_list:
        if user_interest.lower() in plant["category"].lower():
            recommendations.append(plant["name"])
    return recommendations
```

A1.3.2 SEARCH RANKING ALGORITHM

This module ranks plant search results based on relevance. It prioritizes exact matches and frequently searched plants. The ranking improves accuracy and user experience during plant exploration.

```
# search_ranking.py
def rank_plants(query, plants):
    ranked = sorted(
        plants,
        key=lambda p: query.lower() in p["name"].lower(),
        reverse=True
    )
    return ranked
```

A1.3.3 DATA PROCESSING MODULE

This module processes plant data before sending it to the frontend. It filters unnecessary fields and formats output. The module ensures efficient data handling and reduces response size.

```
# data_processing.py
def process_plant_data(plant):
    return {
        "name": plant["name"],
        "uses": plant["medicinalUses"],
        "category": plant["category"]
    }
```

A1.4 DATABASE SCHEMA

A1.4.1 PLANT SCHEMA

This schema defines the structure for storing plant-related data in MongoDB. It includes fields such as name, category, medicinal uses, and image path. The schema ensures structured and efficient storage of plant information.

```
// plantModel.js
const mongoose = require("mongoose");
```

```
const plantSchema = new mongoose.Schema({
  name: String,
  category: String,
  medicinalUses: String,
  image: String
});
module.exports = mongoose.model("Plant", plantSchema);
```

A1.4.2 USER INTERACTION SCHEMA

This schema stores user interactions such as bookmarks and notes. It helps track user activity and improves personalization features. The schema enhances user experience by enabling saved preferences.

```
// userInteraction.js
const mongoose = require("mongoose");
const schema = new mongoose.Schema({
  userId: String,
  plantId: String,
  notes: String,
  bookmarked: Boolean
});
module.exports = mongoose.model("UserInteraction", schema);
```

APPENDIX II - IMPLEMENTATION AND RESULTS

The following screenshots present the key interfaces and functional screens of the deployed Virtual Herbal Garden platform, demonstrating the system's complete implementation across all major modules.

A2.1 HOME PAGE

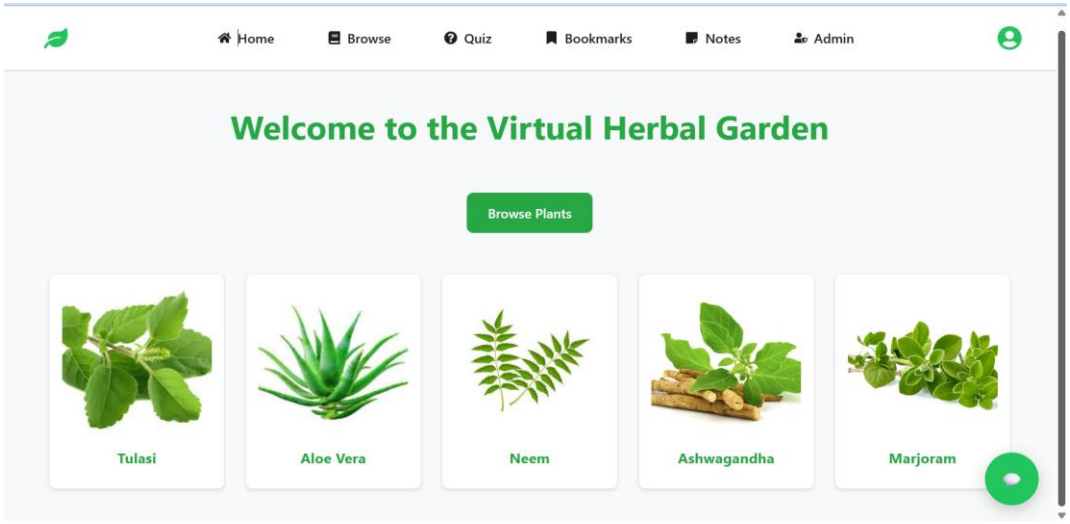

Figure A2.1: Home Page - Welcome to Virtual Herbal Garden

The home page of the Virtual Herbal Garden presents a clean, welcoming interface with a prominent navigation bar offering access to Home, Browse, Quiz, Bookmarks, Notes, and Admin sections. The central welcome message invites users to explore the platform's plant library through a prominent 'Browse Plants' call-to-action button. A horizontal gallery immediately below showcases featured medicinal plants including Tulasi, Aloe Vera, Neem, Ashwagandha, and Marjoram with thumbnail images and common names, providing immediate visual engagement and entry points to the plant library.

A2.2 BROWSE PLANTS PAGE

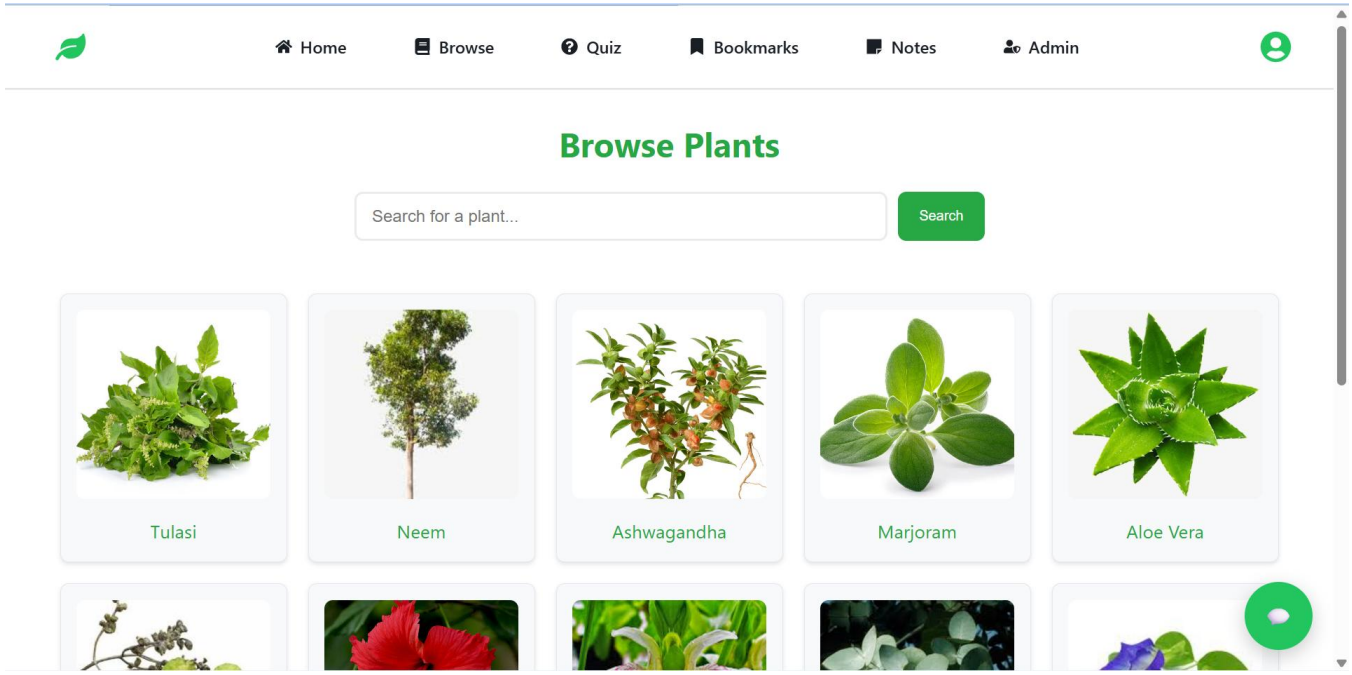

Figure A2.2: Browse Plants Page with Search and Filtering

The Browse page displays the complete plant library in a responsive grid layout with thumbnail cards for each plant. The search bar at the top accepts keyword queries with real-time results updating as the user types. Category filter panels on the left sidebar allow filtering by plant type, AYUSH system, medicinal application, and plant part used. Each plant card shows the plant's common name, scientific name, and primary category tags, providing sufficient context for users to identify relevant plants without opening individual detail pages.

A2.3 PLANT DETAILS PAGE

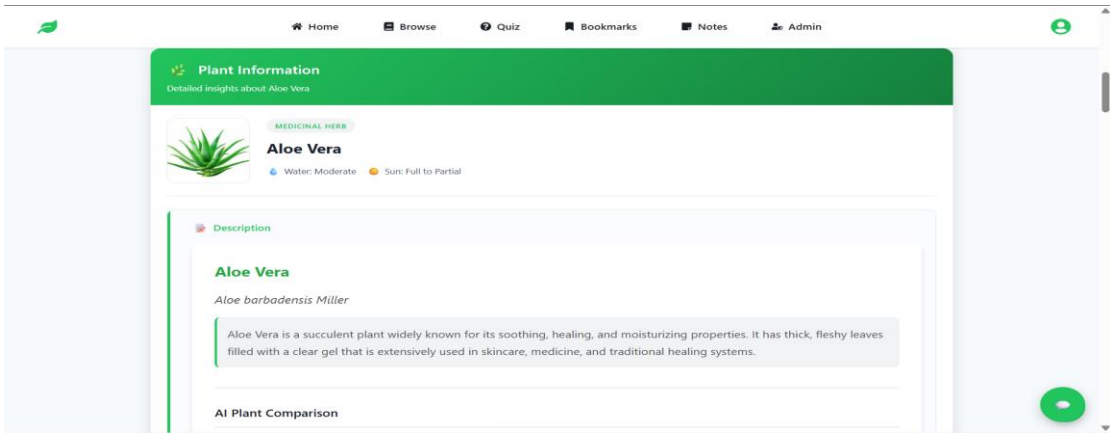

Figure A2.3: Specific Plant Details Page

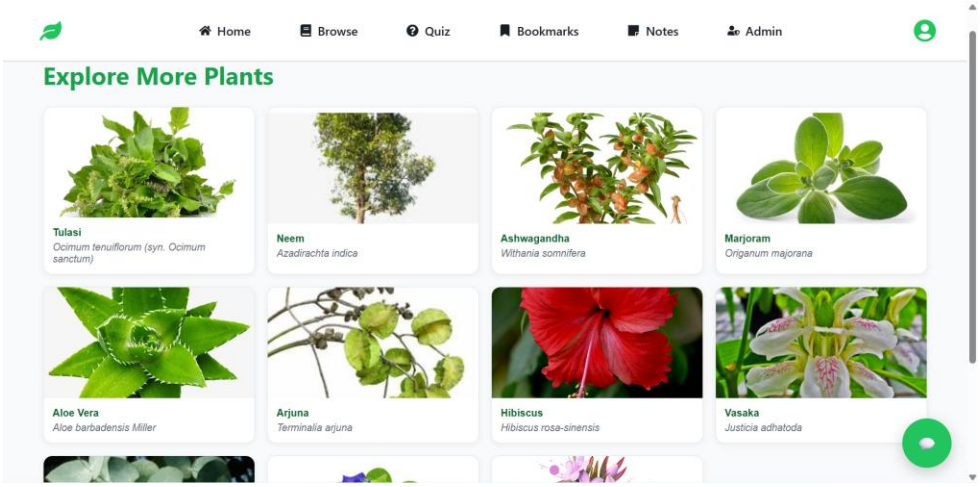

Figure A2.4: Plant Information Tabs View

The Plant Information page provides a comprehensive display of botanical data organized into clearly labeled tabs. The shown example for Aloe Vera (Aloe barbadensis Miller) displays the scientific name, synonym, ideal temperature range, and water requirement at the top. The Description section provides a concise botanical overview. Below the main description, expandable sections for Taxonomy, Morphology, Phytochemistry, Medicinal Properties, Cultivation, and Safety Precautions allow users to access specific information

categories without scrolling through the entire content at once. The bookmark and notes action buttons are prominently placed at the top of the information panel for easy access.

A2.4 3D PLANT VISUALIZATION

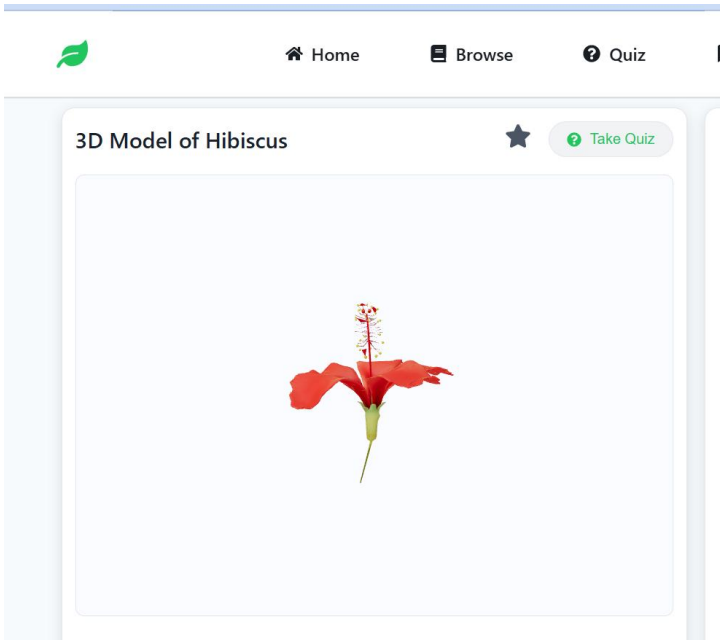

Figure A2.5: Interactive 3D Model of Specific Plant

The 3D Visualization panel demonstrates the platform's core innovative feature. The Aloe Vera model is displayed in a naturalistic scene with appropriate lighting that highlights the plant's characteristic geometry. The model shown demonstrates the interactive state during a rotation operation, with the plant model rotated to show the structural detail of the leaf arrangement from a non-standard angle. The control hint below the scene ('Drag to rotate | Scroll to zoom') guides new users on interaction. The 'Add Note' button in the top right corner of the panel enables inline note creation without navigating away from the 3D view.

A2.5 NOTES SECTION

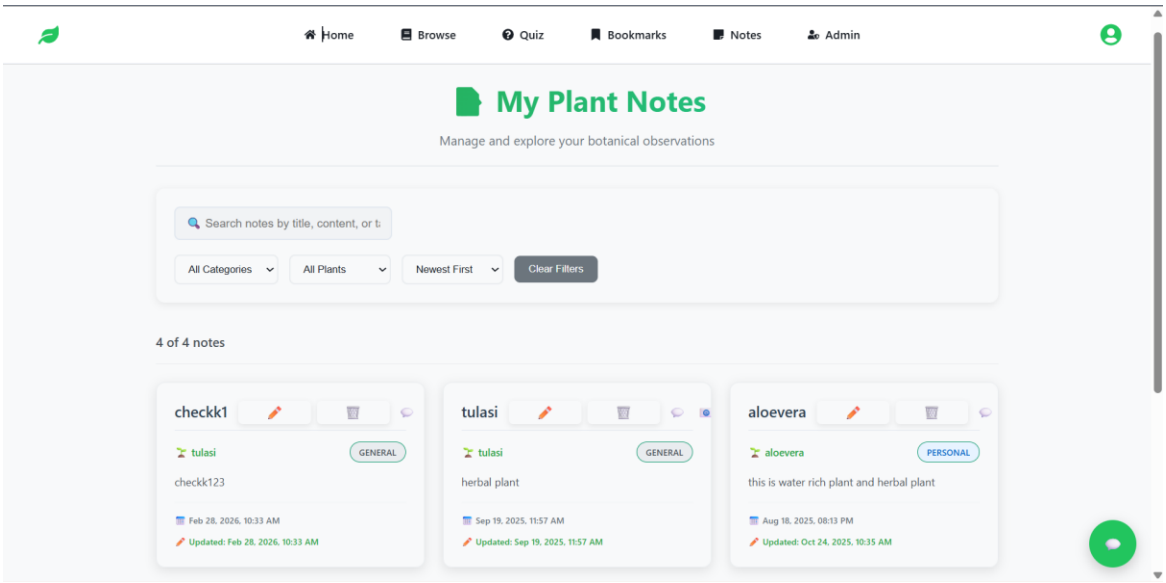

Figure A2.6: Notes Section - Personal Botanical Observations

The Notes Section provides users with a personal knowledge management workspace for botanical observations. The interface displays a searchable, filterable list of existing notes organized by plant association. Each note card shows the plant name, a note category tag, the note content preview, and creation and modification timestamps. The filtering controls allow notes to be filtered by plant category, time period, and tags, enabling users to quickly locate specific observations within a large collection of personal notes. The 'Add Note' button opens an inline note creation form above the existing notes list.

A2.6 BOOKMARKS PAGE

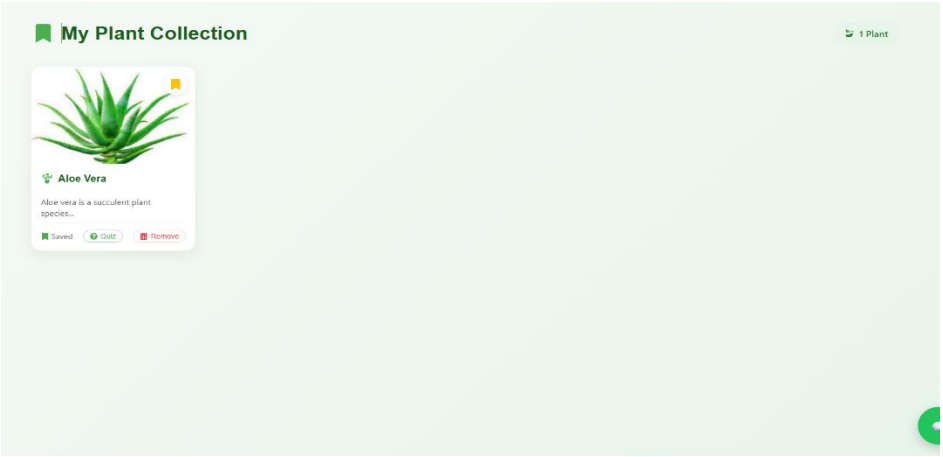

Figure A2.7: Bookmarks Page - Personal Plant Library

The Bookmarks page presents the user's saved plant collection in a card grid layout consistent with the main browse page. Each bookmarked plant card shows the thumbnail, common name, and category tags, with a remove bookmark button accessible via a small icon on the card. The bookmarks page serves as a personalized herbal library where users can curate a collection of plants most relevant to their individual learning focus or professional interest. A search bar enables quick location of specific bookmarked plants within large collections.

A2.7 QUIZ MODULE

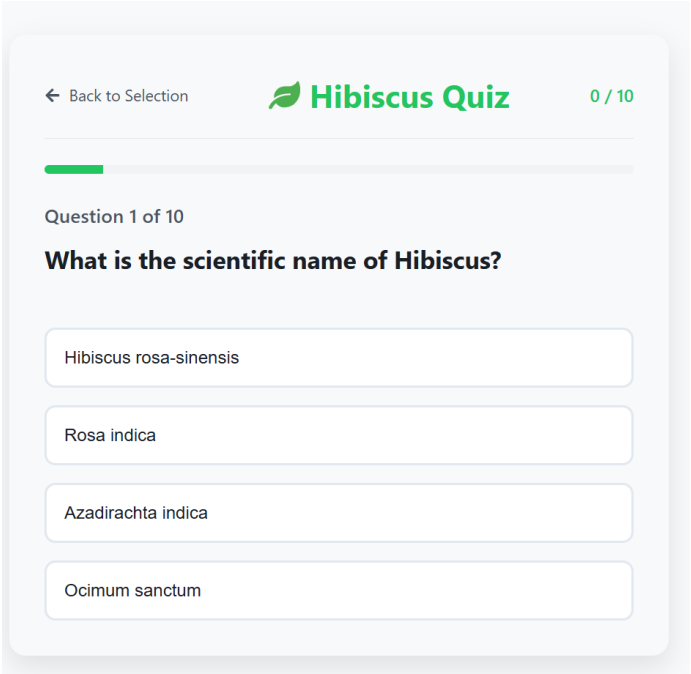

Figure A2.8: Quiz Module - Knowledge Assessment Interface

The Quiz Module interface presents botanical knowledge assessment questions drawn from the plant database. The shown question asks the user to identify a plant based on a 3D model screenshot with four multiple-choice options provided. A progress indicator at the top shows the current question number within the assessment sequence. The interface presents questions with clear visual hierarchy, adequate spacing between options, and an immediate Submit button. Correct and incorrect answers are revealed immediately after submission with explanatory feedback linking to the relevant plant information.

A2.8 ADMIN PANEL

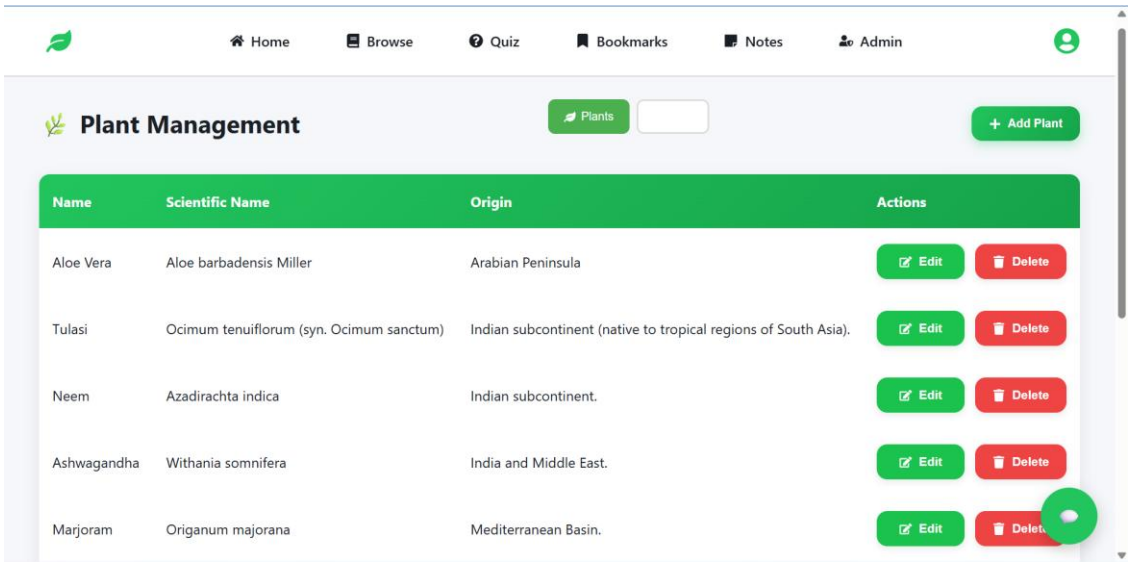

Figure A2.9: Admin Panel - Content Management Interface

The Admin Panel provides the platform administrators with comprehensive content management capabilities. The panel displays a full list of all plant records with their publication status, view counts, and management action buttons for edit, delete, and toggle publish status. The 'Add New Plant' button opens a comprehensive multi-section form for entering all botanical information categories. The admin interface also provides access to user management, system analytics dashboards, and quiz question management. The statistics shown at the top of the admin panel display total plants, published plants, total users, and total quiz attempts.

A2.9 SYSTEM PERFORMANCE RESULTS

METRIC	10 USERS	50 USERS	100 USERS	STATUS
Avg. API Response (ms)	45	68	91	PASS
Plant List Endpoint (ms)	38	62	78	PASS
Plant Detail Endpoint (ms)	52	89	118	PASS
Search Endpoint (ms)	28	41	48	PASS
3D Model Load Time (s)	1.2	1.6	1.9	PASS
Error Rate (%)	0.00	0.00	0.02	PASS

Table A2.1: System Performance Under Varying Loads

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LIST OF PUBLICATIONS

International Conference Papers

[1] Sakthi S, Sanjay King C, Shriram N, Nithilan S, and Ms. P. Raja Packiam, *"Virtual Herbal Garden: Bridging Traditional Knowledge and Technology"* - presented at the International Conference on Computational Intelligence and Industry 5.0 (ICCII), 2026, and certificates of participation have been awarded.